ReligionBERT: Domain-Adaptive Pre-Training of BERT on Biblical Corpora for Religious NLP Tasks

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Abstract

Pre-trained language models such as BERT achieve strong performance on general NLP benchmarks but struggle with the distinctive vocabulary, archaic syntax, and dense theological symbolism found in religious texts. This paper presents ReligionBERT and MultiReligionBERT, two domain-adapted BERT models produced by continued masked language modelling pre-training on the Bible corpus across monolingual English and 12-language multilingual settings. Alongside these models, three automatically constructed task-specific datasets are introduced: a 21,994-pair semantic similarity dataset, a 7,726-sample 66-class book classification dataset, and a 1,199-example LLM-assisted extractive question answering dataset verified at 100% quality on a 200-sample human review. Benchmarking across four model variants and three task types shows that ReligionBERT outperforms generic BERT on five of six metrics, with the strongest gain of +7.50 Exact Match points on extractive QA. Perplexity on held-out Bible text falls from 15.48 for generic BERT to 3.73 for ReligionBERT, a 75.9% reduction. Zero-shot cross-lingual transfer on five African languages demonstrates that multilingual domain-specific pre-training confers measurable advantages over mBERT, including a result on Ewe where mBERT scores 0.00% accuracy and MultiReligionBERT scores 3.37%. Both models and all datasets are released publicly on HuggingFace.

Keywords: domain-adaptive pre-training, BERT, biblical corpus, religious NLP, cross-lingual transfer, African languages, extractive QA

1. Introduction

The rapid advancement of pre-trained language models (PLMs) has fundamentally transformed the field of natural language processing (NLP), enabling systems to achieve near-human or superhuman performance on a wide range of language understanding and generation benchmarks [1, 2] . At the core of this progress is the insight that models trained on massive, general-purpose corpora, such as Wikipedia, BooksCorpus, and CommonCrawl, can acquire transferable linguistic representations that benefit diverse downstream tasks through fine-tuning. Models such as BERT [2], RoBERTa [3], and their multilingual counterparts have established strong baselines across tasks including text classification, named entity recognition, question answering, and natural language inference.

However, the general-purpose nature of these corpora also constitutes a fundamental limitation. The distributional properties of Wikipedia and web-crawled text reflect the vocabulary, syntax, and conceptual structures of contemporary, secular, encyclopedic writing. When applied to specialised domains whose language departs significantly from these norms, general-purpose PLMs frequently underperform, misinterpret domain-specific terminology, and fail to capture nuanced contextual relationships that domain experts would consider elementary [4]. This limitation has motivated the development of domain-adaptive pre-training (DAPT), a paradigm in which a PLM undergoes additional pre-training on in-domain text before task-specific fine-tuning, effectively recalibrating the model's internal representations toward the target domain.

Religious texts, and biblical literature in particular, represent one of the most linguistically distinctive and socially significant domains in human writing. The Bible, as one of the most translated and widely read texts in human history, exhibits a range of features that sharply distinguish it from general-purpose corpora. These include archaic grammatical constructions rooted in Hebrew, Aramaic, and Koine Greek source languages; dense metaphor and allegory operating across multiple levels of interpretation simultaneously; highly domain-specific theological vocabulary, including terms such as atonement, sanctification, eschatology, and propitiation, whose meanings are not recoverable from general corpus statistics; and long-range intertextual dependencies in which the meaning of a passage in one book is illuminated by or contingent upon passages in entirely different books written centuries apart [5]. When applied to biblical text without domain adaptation, BERT-class models systematically misinterpret theological symbols, conflate semantically distinct religious concepts, and underperform on tasks that require the kind of contextual understanding that domain-adapted models can provide.

The significance of religious NLP extends well beyond academic interest. Religious texts are foundational to the spiritual, cultural, ethical, and intellectual traditions of billions of people worldwide. Applications of NLP to religious corpora include automated biblical commentary generation, theological question answering systems for educational and pastoral use, semantic search and concordance tools for scripture study, cross-lingual alignment of parallel religious translations for low-resource language research, and sentiment and discourse analysis of religious communities on social media. Each of these applications stands to benefit substantially from models that have been adapted to understand the specific linguistic patterns of the domain.

Furthermore, the Bible corpus has a unique structural property that makes it particularly well-suited to NLP research: it is one of the most consistently formatted and widely translated parallel corpora in existence, with full or partial translations available in more than 2,000 languages, including many African languages for which general-purpose NLP resources remain extremely scarce [6]. The verse-based structure of the Bible, in which each verse is an independently addressable unit of text with a unique identifier, provides natural alignment anchors for multilingual tasks and enables the systematic construction of structured datasets without the need for expensive manual annotation.

Despite these compelling motivations, the application of domain-adaptive pre-training to religious corpora has received almost no systematic attention in the NLP literature. Existing work has largely been limited to descriptive analyses of biblical text using classical NLP methods, keyword search, topic modelling, and lexicon-based sentiment analysis, without leveraging the representational power of modern transformer-based PLMs [7]. The growing body of work applying LLMs to religious question answering has primarily used general-purpose models without domain adaptation, leaving open the question of whether specialised pre-training on religious corpora produces measurable and consistent improvements [8].

This study addresses the gaps directly and contributes to knowledge as follows:

- ReligionBERT and MultiReligionBERT are produced via continued masked language modelling pre-training on monolingual English and 12-language multilingual Bible corpora, respectively, and both are released publicly on HuggingFace.
- Three automatically constructed datasets are introduced for religious NLP: a 21,994-pair semantic similarity dataset with structured score assignment, a 7,726-sample 66-class book classification dataset with corrected XML book code mappings, and a 1,199-example LLM-assisted extractive QA dataset verified at 100% quality on a 200-sample human review.
- The first systematic multi-task, multi-model benchmarking study of Bible corpus domain adaptation is provided, covering four model variants across three task types and demonstrating that domain-adapted models consistently outperform their generic counterparts, with a peak gain of +7.50 Exact Match points on extractive QA.

- The first empirical study of zero-shot cross-lingual transfer for African language religious NLP using domain-adapted multilingual models is presented, evaluating on five African languages, including the severely low-resource Ewe, for which standard multilingual models score 0.00% accuracy while MultiReligionBERT achieves 3.37%.

The remainder of this paper is organized as follows. Section 2 reviews related work in domain-adaptive pre-training, multilingual modelling, and NLP for religious texts. Section 3 describes the methodology, including corpus construction, dataset design, model architecture, pre-training configuration, and fine-tuning procedure. Section 4 presents result across all tasks and the cross-lingual transfer experiments. Section 5 provides discussion, analysis of findings, and concluding remarks.

2. Literature Review

2.1 Domain-Adaptive Pre-Training

The concept of transfer learning in NLP predates the transformer era, with early work demonstrating that representations learned on one task could be transferred to benefit related tasks through fine-tuning [9]. The introduction of the transformer architecture [10] and the subsequent development of large-scale masked language models such as BERT [2] dramatically expanded the scope and effectiveness of transfer learning in NLP by enabling the pre-training of deep bidirectional representations that capture rich syntactic and semantic structure from unlabelled text.

A critical insight that emerged from early applications of BERT was that the general-purpose pre-training corpora, while large, do not adequately represent the vocabulary and linguistic patterns of specialised domains. This motivated the development of domain-specific pre-training strategies.

Howard and Ruder introduced ULMFiT, which demonstrated that sequential pre-training on domain-specific corpora followed by task-specific fine-tuning produced consistent gains across text classification tasks, establishing the basic DAPT pipeline that subsequent work would extend to larger models [9].

The most systematic investigation of DAPT was conducted by Gururangan et al. [4], who showed across four domains (biomedical, computer science, news, and reviews) and eight classification tasks that continued pre-training on in-domain text consistently improves downstream performance, regardless of whether the model is subsequently fine-tuned on in-domain or out-of-domain task data. Their work also introduced task-adaptive pre-training (TAPT), in which the model is further pre-trained on the specific unlabelled data associated with a downstream task, yielding additional gains. The DAPT framework established in this work directly informs the methodology of the present study.

BioBERT [11] applied the DAPT paradigm to biomedical literature, pre-training BERT on PubMed abstracts and PubMed Central full-text articles and demonstrating substantial improvements on biomedical named entity recognition, relation extraction, and question answering. SciBERT [1] extended the approach to scientific text more broadly, training from scratch on a corpus of 1.14 million papers from Semantic Scholar across computer science and biomedical domains. LegalBERT [12] demonstrated similar gains in the legal domain, pre-training on legislation, court cases, and legal articles. These studies collectively established DAPT as the standard approach for improving PLM performance on specialised text and provided the methodological foundation upon which the present work builds.

Domain-specific fine-tuning with targeted corpora has also been validated as a complementary strategy to continued pre-training. When a pre-trained model is further trained on a domain corpus, it incorporates domain-specific terminologies, stylistic features, and context-sensitive meanings absent from general training data [13]. Empirical studies confirm that models fine-tuned on domain corpora consistently outperform general models in specialised settings including biomedical text mining [14], financial forecasting [15] , and theological text analysis [16]. However, such fine-tuning carries risks of catastrophic forgetting, in which the model loses general linguistic competence after prolonged exposure to narrow-domain data [17], as well as overfitting when annotated in-domain data is scarce. Approaches such as gradual unfreezing or mixed fine-tuning, supplementing domain data with a proportion of general-domain text, have been proposed to mitigate these effects [13].

Despite the breadth of domains addressed by existing DAPT research, religious text has been absent from systematic investigations of this kind. The linguistic properties of religious text, including archaic syntax, dense theological symbolism, and long-range intertextual dependencies, make it a particularly challenging and interesting domain for DAPT, and the multilingual nature of the Bible corpus creates opportunities for cross-lingual transfer research that do not exist for most specialised domains.

2.2 Multilingual Models and Cross-Lingual Transfer

The development of multilingual pre-trained language models has opened new possibilities for NLP in low-resource languages by enabling zero-shot and few-shot cross-lingual transfer. Multilingual BERT [2], trained jointly on Wikipedia text from 104 languages using a shared subword vocabulary of 119,547 tokens, demonstrated that a single model could achieve reasonable performance on tasks in languages not seen during task-specific fine-tuning, a finding that was initially surprising given that no explicit cross-lingual alignment objective was used during pre-training.

Subsequent analysis of mBERT's cross-lingual capabilities revealed that the shared multilingual vocabulary and joint training objective encourage the model to develop language-agnostic representations for concepts that appear in similar syntactic and semantic contexts across languages. However, mBERT's cross-lingual transfer performance varies significantly across language pairs, with typologically similar languages benefiting more from transfer than typologically distant ones.

XLM-RoBERT [18] substantially improved on mBERT by adopting the RoBERTa pre-training recipe, removing the next sentence prediction objective, training with larger batches and more data, and scaling the multilingual corpus to 2.5 terabytes of CommonCrawl text in 100 languages. The larger and more diverse pre-training corpus improved both monolingual and cross-lingual performance, particularly for low-resource languages that are underrepresented in Wikipedia.

For African languages specifically, cross-lingual transfer remains a significant challenge due to the extreme scarcity of NLP resources for most African languages, the typological diversity of the African language families, and the underrepresentation of African languages in the CommonCrawl and Wikipedia corpora that serve as the basis for most multilingual pre-training. Languages such as Ewe, a Gbe language spoken primarily in Ghana and Togo, have extremely limited digital text resources and are barely represented in standard multilingual models. The Bible corpus, which includes translations in many African languages, represents one of the most consistent and systematically formatted multilingual text resources available for African NLP, making it a particularly valuable pre-training corpus for cross-lingual transfer to African language tasks.

2.3 NLP for Religious and Biblical Texts

Computational approaches to religious texts have a long history in digital humanities, beginning with concordance tools and keyword-in-context analysis that enabled systematic exploration of lexical patterns across scripture. The advent of statistical NLP methods extended the toolkit available to computational biblical scholars to include topic modelling, sentiment analysis, and distributional semantic representations.

The Bible corpus has been used in computational linguistics as a benchmark for machine translation and cross-lingual alignment due to its extensive multilingual coverage and parallel verse structure [6]. Naude and Miller-Naude [6] documented the use of parallel Bible corpora in minority language research, noting that Bible translations often constitute the only substantial digitised text resource available for many endangered and low-resource languages, making them invaluable for both linguistic documentation and NLP research.

Nandan et al. conducted a comparative analysis of the Bible, Quran, and Bhagavad Gita using NLP approaches, applying topic modelling, sentiment analysis, and cross-textual semantic similarity measures to characterise the linguistic and thematic properties of these major religious corpora. Their work highlighted the distinctive vocabulary distributions of religious texts and the challenges they pose for general-purpose NLP tools.

More recent work has explored the application of large language models to religious question answering and theological reasoning. Zhong et al. [8] examined the opportunities and challenges of applying LLMs to low-resource languages in humanities research, including religious scholarship, and identified domain-specific pre-training as a key direction for improving performance. However, no prior work has implemented and systematically evaluated continued BERT pre-training on biblical corpora, constructed task-specific benchmarks for religious NLP, or investigated zero-shot cross-lingual transfer for African language religious NLP.

2.4 Interpretability of Domain-Adapted Models

Understanding why domain-adapted models outperform their generic counterparts is an important complement to empirical benchmarking. Alain and Bengio [9] introduced linear probing classifiers as a tool for investigating what information is encoded at each layer of a deep neural network: by training a simple logistic regression classifier on frozen layer representations, one can assess the extent to which each layer encodes linearly separable information relevant to a given task. Applied to domain-adapted models, probing provides direct mechanistic evidence for whether adaptation genuinely shifts internal representations toward domain-relevant features or merely produces surface-level vocabulary matching.

Vig [19] developed multiscale attention visualisation tools for transformer models, enabling the visualisation of attention patterns at the level of individual heads, layers, and token pairs. Attention visualisation has been used to investigate which token relationships models attend to when processing domain-specific text, providing qualitative evidence for the kinds of contextual associations that domain adaptation reinforces.

In the context of this study, probing classifier experiments conducted on the ReligionBERT model reveal that domain adaptation produces measurable improvements in the encoding of religious categorical distinctions at all representational depths, with the largest gain of 7.12 percentage points on theological section classification at the final transformer layer. This finding, reported in detail in a companion paper, confirms that the performance gains observed in downstream task benchmarking reflect genuine shifts in internal representations rather than surface-level statistical artefacts.

Table 1: Comparative summary of related studies in domain adaptation for specialised text corpora.

2.5 Parameter-Efficient Fine-Tuning

The computational cost of full fine-tuning, which updates all model parameters for each downstream task, is prohibitive when working with limited hardware or across multiple task domains [20]. This has motivated growing interest in Parameter-Efficient Fine-Tuning (PEFT) methods, which adapt a model by updating only a small fraction of parameters while keeping most of the pre-trained backbone frozen. PEFT approaches are grounded in the observation that large language models encode rich general-purpose representations during pre-training, and that effective task adaptation does not require full parameter re-optimisation [21].

Among the most widely used PEFT techniques are adapter modules, prefix-tuning, LoRA (Low-Rank Adaptation), and prompt-tuning. Adapter-based methods insert small trainable neural layers into transformer architectures while freezing the backbone. Prefix-tuning and prompt-tuning optimise task-specific soft prompts prepended to model inputs while leaving model weights unchanged [22]. LoRA injects low-rank trainable matrices into the model's weight updates, drastically reducing the number of trainable parameters and making it particularly attractive for resource-constrained environments [23]. Benchmarking studies have found that LoRA and prefix-tuning achieve accuracy comparable to full fine-tuning while reducing trainable parameters by over 90% [20].

In the context of domain-specific corpora such as religious or legal texts, PEFT methods are especially valuable because such domains often lack extensive annotated datasets, and PEFT enables adaptation using smaller amounts of training data while preserving the pre-trained model's broad linguistic competence [21, 24]. Recent developments increasingly combine PEFT with domain-specific continued pre-training, a hybrid strategy that reduces computational costs while retaining the performance benefits of domain adaptation [25].

2.6 Domain Resources for Religious Corpora

Religious texts are among the most widely read and historically significant documents, forming the foundation of spiritual, cultural, and intellectual traditions across the world [26] . Their preservation makes them invaluable for NLP research, particularly in exploring linguistic diversity, semantic depth, and long-sequence dependencies. However, compared to general-purpose corpora such as Wikipedia or Common Crawl, religious corpora remain limited in availability, structure, and accessibility.

A number of religious corpora have been digitised and annotated for computational research. The Bible Corpus exists in hundreds of translations across diverse languages, making it one of the most multilingual parallel corpora available [6, 27]. Its verse-based structure is particularly useful for machine translation, semantic similarity, and cross-lingual alignment. The Quranic Arabic Corpus provides morphological, syntactic, and semantic annotation layers for the Quran [28, 29], making it a critical resource for studying morphologically rich and classical languages. The Hadith Corpus, consisting of narrations attributed to the Prophet Muhammad, presents research challenges due to its complex isnad (chain of narration) structures and classical Arabic prose [30].

Beyond the Abrahamic traditions, the Bhagavad Gita Corpus and other Sanskrit texts offer study of ancient Indian literature [7, 31] , where inflectional morphology, compounding, and context-rich semantics challenge standard NLP techniques. Initiatives including the Pali Canon Digital Project and the Chinese Buddhist Canon digital corpus make Buddhist texts accessible for linguistic and semantic analysis [32, 33]. Collaborative projects such as OpenITI (Open Islamicate Texts Initiative) and Sefaria further expand access to Islamicate and Jewish texts with integrated commentaries and cross-references, supporting the modelling of intertextuality and diachronic language change [34] .

Despite these advances, several gaps persist. Coverage remains uneven: while Abrahamic scriptures are extensively digitised, many African traditional religions, indigenous oral traditions, and lesser-studied faiths remain underrepresented in corpus form. Issues of translation bias complicate corpus development, since religious texts rely on metaphor, symbolism, and doctrinal interpretation that may vary significantly across sects and denominations. Ethical and cultural sensitivity is paramount, as inappropriate or biased curation can lead to misrepresentation of deeply held beliefs [5]. These challenges emphasise the importance of collaborative, interdisciplinary approaches involving computational linguists, theologians, and cultural experts.

2.7 Long-Sequence Modelling

Traditional transformer-based language models such as BERT and GPT are designed with fixed input lengths, typically 512 tokens, which presents a serious limitation when dealing with long religious texts such as the Bible, Quran, Vedas, or Buddhist scriptures [35, 36] . These texts often contain narratives, commentary, and layered references that extend beyond short passages, making it difficult for models to capture context across chapters or books. Truncation of text leads to loss of semantic coherence, especially in domains that rely on intertextuality and cross-references [37]. Furthermore, the quadratic complexity of self-attention means that processing thousands of tokens becomes resource-intensive, requiring specialised architectures or approximations [38].

Recent research has addressed these limitations through innovations in architecture and training. Models such as Longformer, BigBird, and Performer introduce sparse or linearised attention mechanisms that allow efficient processing of thousands of tokens [39, 40] . Hierarchical models have also been proposed to better represent multi-level structures in texts, aligning local sentence-level understanding with global thematic structures such as doctrine, parables, or moral reasoning spread across chapters [41]. Parameter-efficient tuning and domain-specific pre-training further make it possible to adapt long-sequence models to religious corpora without excessive computational resources [25].

Authors	**Domain**	**Method**	**Key Finding**	**Gap Addressed Here**
Gururangan et al. [4]	General	DAPT on domain corpora	Consistent gains across 4 domains	Religious domain; cross-lingual
Lee et al. [11]	Biomedical	BioBERT pre-training	Strong biomedical NLP gains	Religious setting; African languages
Beltagy et al. [1]	Scientific	SciBERT from scratch	State-of-the-art scientific tasks	Multilingual; low-resource
Chalkidis et al. [12]	Legal	LegalBERT pre-training	Improved legal NLP tasks	Religious domain; multilingual
Conneau et al. [18]	Multilingual	XLM-RoBERTa large-scale	Strong cross-lingual transfer	Domain-specific religious adaptation

Authors	Domain	Method	Key Finding	Gap Addressed Here
Nandan et al. [7]	Religious texts	Keyword and topic analysis	Descriptive insights on Bible, Quran	BERT adaptation; benchmarking
Naudé & Miller-Naudé [6]	Bible / minority langs	Parallel corpus survey	Bible translations are the primary digital resource for many low-resource languages	BERT adaptation; task benchmarking
Wang et al. [14]	Biomedical	Continued pre-training survey	In-domain pre-training improves multiple benchmarks even with modest data	Religious domain; African languages
This study	Religious (Bible)	DAPT + multi-task + probing	Gains on 5/6 metrics; African transfer	All of the above

3. Methodology

This section describes the complete methodological framework of the study, organised into five components: pre-training corpus construction, fine-tuning dataset construction, base model selection and architecture, continued pre-training configuration, and fine-tuning procedure. Figure 1 provides an overview of the dataset construction pipeline, and Figure 3 illustrates the full experimental pipeline from corpus collection through evaluation.

3.1 Computational Infrastructure

3.1.1 Hardware Configuration

The T4 GPU was used for the following tasks: all initial data exploration and preprocessing scripts, tokenization of the English and multilingual training corpora, the 1,000-step pilot training run used to validate the pipeline, the initial continued pre-training runs for both the English and multilingual models across multiple resumed sessions, all dataset construction scripts for the semantic similarity, classification, and question answering datasets, and the full interpretability analysis including hidden state extraction, probing classifier training, and attention visualisation.

The A100 GPU, accessed via paid compute units, was used for the following tasks: the fine-tuning of all four model variants on all three downstream tasks (eight fine-tuning runs across semantic similarity and classification, and four runs for question answering), the XLM-RoBERTa fine-tuning for the cross-lingual baseline, and the cross-lingual zero-shot evaluation runs. The decision to use A100 units for fine-tuning was driven by two factors. First, the T4's 12-hour session limit made it difficult to complete longer fine-tuning jobs, particularly for the multilingual models, which have 178 million parameters versus 109 million for the English models. Second, the A100's significantly higher throughput reduced fine-tuning times from several hours to under an hour for most tasks, which made iteration and re-running easier when hyperparameters needed adjustment.

A practical challenge related to storage was encountered and resolved during the continued pre-training phase. The HuggingFace Trainer saves deleted checkpoints to the Google Drive Trash rather than permanently removing them. Because Drive Trash items continue to count toward the user's storage quota, this caused a rapid and unexpected accumulation of storage usage across multiple training sessions, reaching approximately 211 GB before the issue was identified. A custom TrainerCallback was implemented that uses Python's shutil.rmtree function to permanently delete old checkpoint directories, bypassing the Drive Trash entirely.

This intervention kept storage usage stable throughout subsequent training runs. A further incident occurred in which both completed pre-training models were accidentally deleted from Google Drive during an attempt to free storage by removing large files. Following this incident, a mandatory procedure was established to push all trained models to the HuggingFace Hub immediately upon completion of training, providing a permanent off-site backup independent of Google Drive.

3.1.2 Software Stack

The following software versions were used throughout the project. Python 3.12 was the runtime environment for all scripts and notebooks. HuggingFace Transformers version 5.0.0 was used for all model loading, training, and evaluation. HuggingFace Datasets was used for dataset loading and processing. PyTorch with CUDA support was used as the deep learning backend with 16-bit floating point mixed precision training enabled throughout. WandB version 0.25.1 was used for experiment tracking and live loss curve visualisation. The Groq Python client was used for the QA dataset generation pipeline. Matplotlib and scikit-learn were used for result visualisation and probing classifier training, respectively. All trained models were published to the HuggingFace Hub for permanent backup and accessibility.

Several breaking changes were encountered in Transformers 5.0.0 that caused failures in the training pipeline. These are documented here for reproducibility. The TrainingArguments parameter evaluation_strategy was renamed to eval_strategy. The tokenizer argument in the Trainer class was renamed to processing_class. The logging_dir argument was deprecated and had to be removed. The save_steps value was required to be an exact integer multiple of eval_steps when load_best_model_at_end was set to True, otherwise a ValueError was raised at startup.

The model checkpoint format changed such that LayerNorm parameters previously named gamma and beta were renamed to weight and bias in newer PyTorch versions, producing harmless missing and unexpected key warnings on checkpoint load that did not affect training correctness. Additionally, the model downloaded from the HuggingFace Hub for the initial baseline had to be specifically loaded as BertForMaskedLM rather than the base BertModel class, since the latter lacks the Masked Language Modelling prediction head required for continued pre-training.

3.2 Pre-Training Corpus

The primary corpus was sourced from the christos-c/bible-corpus GitHub repository, which provides 108 XML-formatted Bible translations with consistent verse-level structure. The repository was selected because its raw XML format exposes verse-level metadata through a structured identifier in the format b.BOOK.CHAPTER.VERSE, which could be used directly to construct classification and evaluation labels without additional annotation. All verse text was extracted from the seg XML element, preserving the book and chapter hierarchy.

For monolingual pre-training (ReligionBERT), the King James Version and the World English Bible were selected and concatenated, yielding 62,197 verses totalling approximately 7.78 MB of text. The KJV was selected as the primary English translation due to its archaic syntax and vocabulary, which represent the maximum departure from general-purpose pre-training corpora and therefore provide the strongest test of domain adaptation. The WEB was included as a second English translation to provide additional stylistic variation within the monolingual corpus.

For multilingual pre-training (MultiReligionBERT), twelve languages were selected to provide geographic, typological, and script diversity while maintaining adequate per-language verse coverage. The selected languages were English (KJV and WEB), French, Spanish, Portuguese, German, Amharic, Shona, Xhosa, Malagasy, Somali, and Zarma, producing a combined corpus of 372,652 verses. Two additional languages with New Testament-only translations, Ewe and Swahili, were excluded from pre-training due to their incomplete corpus coverage but were retained for cross-lingual evaluation. The combined pre-training corpus across both monolingual and multilingual settings comprised 388,448 verses.

A technical challenge was encountered during multilingual tokenisation: processing the full 372,652-verse multilingual corpus in a single pass exceeded the available RAM on Google Colab instances. This was resolved by processing the corpus in eight chunks of approximately 50,000 lines, saving each tokenised chunk as a separate HuggingFace dataset shard and concatenating the shards into a single dataset object before pre-training. The English monolingual corpus was tokenised successfully in a single pass, producing a 54.13 MB tokenised dataset file. Table 2 summarises corpus statistics.

Table 2. Pre-training corpus statistics by setting.

Corpus	Languages	Verses	Base Model
English Bible (KJV + WEB)	1 (English)	62,197	bert-base-uncased

Corpus	Languages	Verses	Base Model
Multilingual Bible	12	372,652	bert-base-multilingual-cased

3.3 Fine-Tuning Dataset Construction

Three task-specific datasets were constructed automatically from the Bible corpus without manual annotation, except for the human verification step applied to the QA dataset. Figure 1 illustrates the full pipeline. All datasets used an 80/10/10 train/validation/test split stratified by category where applicable.

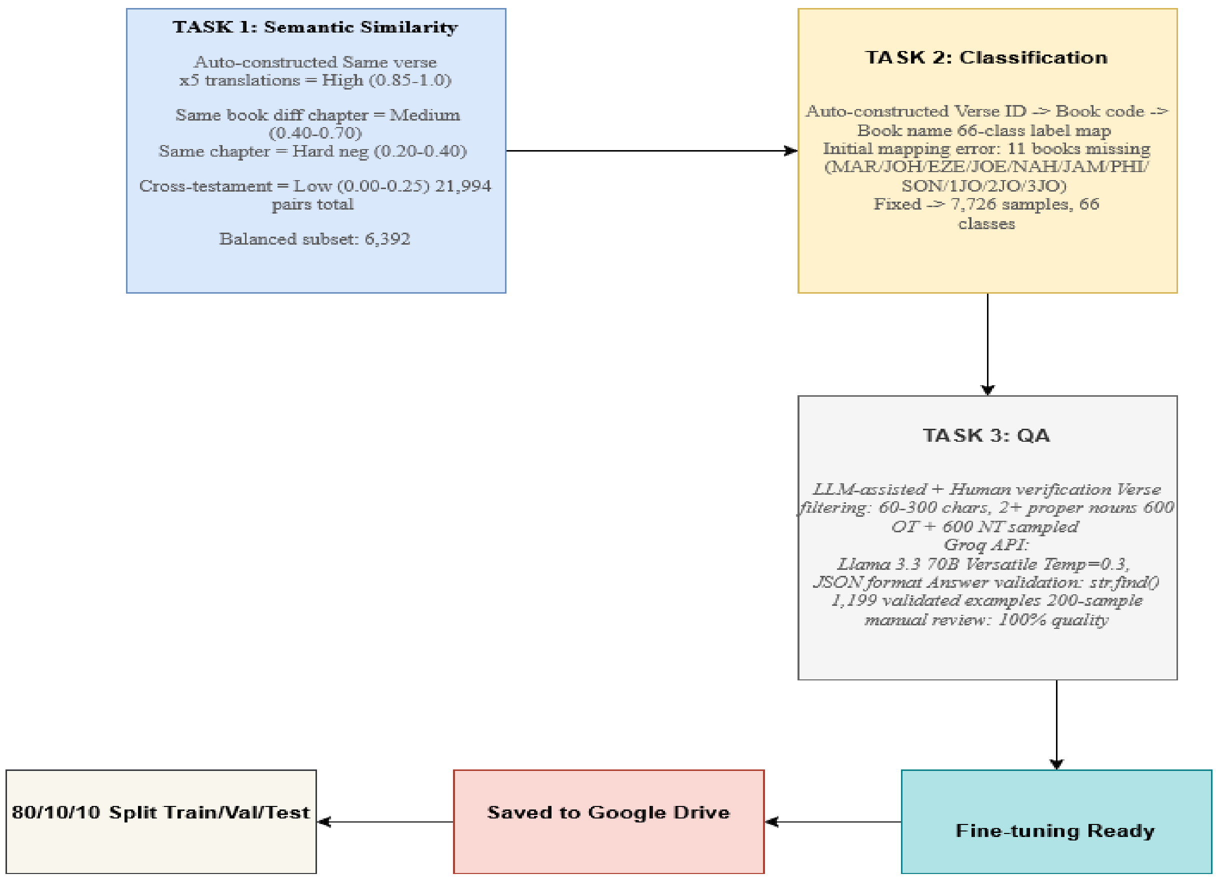

Figure 1. Automatic and LLM-assisted dataset construction pipeline for the three fine-tuning tasks.

3.3.1 Semantic Similarity Dataset.

The semantic similarity dataset was constructed by assigning continuous similarity scores to verse pairs based on the structural properties of the parallel corpus. Four categories of pairs were generated. High-similarity pairs linked the same verse identifier across five cross-translation language pairs and received scores sampled uniformly from 0.85 to 1.00, reflecting the expectation that the same verse expressed in different translations is semantically near-identical. Medium-similarity pairs drew verses from the same book but different chapters, receiving scores from 0.40 to 0.70, reflecting thematic but not propositional similarity. Hard-negative pairs drew verses from the same chapter, receiving scores from 0.20 to 0.40, reflecting incidental proximity without strong semantic overlap. Low-similarity pairs paired Old Testament verses against New Testament verses from thematically distant books, receiving scores from 0.00 to 0.25. The full dataset contained 21,994 pairs, with high-similarity pairs constituting 68.2% of the total. A balanced subset of 6,392 pairs was produced by downsampling the dominant class for use in the first stage of fine-tuning. Table 3 summarises dataset statistics.

Table 3. Semantic similarity dataset statistics.

Category	Pairs	Score Range	Construction Method
High similarity	15,000	0.85-1.00	Same verse, five cross-translation pairs
Low similarity	4,000	0.00-0.25	Cross-testament OT-NT random pairs
Medium similarity	1,554	0.40-0.70	Same book, different chapters
Hard negative	1,440	0.20-0.40	Same chapter, adjacent verses
Full dataset	21,994	0.00-1.00	All categories combined
Balanced subset	6,392	0.00-1.00	Downsampled for class balance

3.3.2 Book Classification Dataset.

The book classification dataset was constructed by labelling each verse with its source book, producing a 66-class classification task. An initial construction run produced only 55 of the 66 expected classes due to eleven non-standard book codes in the XML corpus: MAR was used instead of the standard MRK for Mark, JOH instead of JHN for John, PHI instead of PHP for Philippians, and 1JO, 2JO, 3JO instead of 1JN, 2JN, 3JN for the Johannine epistles, among others.

All eleven non-standard codes were identified and corrected through a custom mapping lookup, and the dataset was regenerated. The corrected dataset contained 7,726 samples across all 66 canonical books of the Protestant Bible, with the Old Testament contributing 4,894 samples and the New Testament contributing 2,832 samples.

3.3.3 Extractive Question Answering Dataset.

No suitable existing extractive QA dataset for biblical text was identified during a literature survey. A dataset of 1,199 question-answer pairs was constructed using Llama 3.3 70B Versatile via the Groq API, with temperature set to 0.3 and response format set to JSON to ensure parseable output. Verse selection filtered the English KJV corpus for passages between 60 and 300 characters in length containing at least two capitalised words after the first word, a heuristic that effectively selects theologically substantive verses containing proper nouns such as names of persons, places, and divine attributes.

Approximately 600 Old Testament and 600 New Testament passages were selected to balance testamental coverage. Each selected verse was submitted to the language model with a structured prompt specifying that the model must generate a factual question answerable from the verse alone, and that the answer string must be a verbatim substring of the verse. Answer validity was enforced programmatically using Python's str.find method with case-insensitive fallback, rejecting any answer string that was not found as an exact substring of the source verse. A 200-sample manual review of the first 200 training examples found zero quality issues, confirming a verified quality rate of 100%. The dataset was formatted in SQuAD v2-compatible JSON format [8]. Table 4 summarises dataset properties.

Table 4. Extractive QA dataset properties.

Property	**Value**
Total examples	1,199
Generation model	Llama 3.3 70B Versatile (Groq API, temperature 0.3)
Validation method	Exact substring matching (Python str.find)
Human review sample	200 examples
Issues found	0 (100% verified quality)
Train / Validation / Test	959 / 120 / 120
OT / NT balance	~600 / ~600 examples
Format	SQuAD v2 compatible JSON

3.4 Base Models and Architecture

Two base models were used. bert-base-uncased [2] is a 12-layer bidirectional transformer encoder with 109 million parameters, a 30,522-token WordPiece vocabulary, a hidden dimension of 768, 12 attention heads per layer, and a feed-forward dimension of 3,072. It was pre-trained on English Wikipedia and BooksCorpus using masked language modelling and next sentence prediction objectives. bert-base-multilingual-cased (mBERT) [2] is a 12-layer encoder with 178 million parameters and a shared 119,547-token multilingual vocabulary, pre-trained on Wikipedia text from 104 languages. XLM-RoBERTa base [18], with approximately 279 million parameters pre-trained on 2.5 terabytes of CommonCrawl text in 100 languages, was included as an additional cross-lingual baseline in the transfer experiments. Domain-adaptive pre-training on the Bible corpus produced ReligionBERT from bert-base-uncased and MultiReligionBERT from bert-base-multilingual-cased. Figure 2 illustrates the BERT architecture.

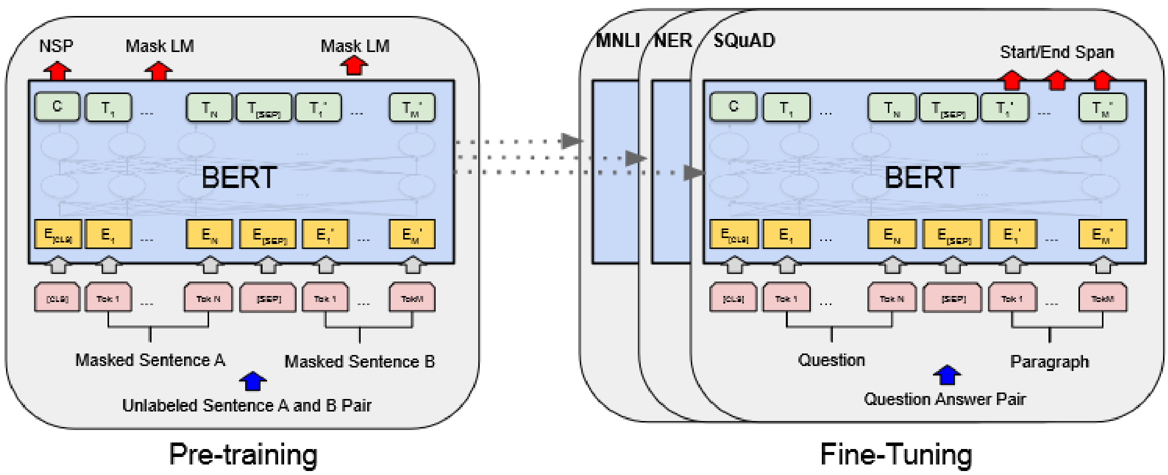

Figure 2. BERT pre-training and fine-tuning architecture showing multi-layer bidirectional Transformer blocks [2].

3.5 Pre-Training Configuration

Both domain-adapted models were trained using the masked language modelling objective with 15% random token masking, following the original BERT masking procedure in which 80% of selected positions are replaced with the [MASK] token, 10% with a random vocabulary token, and 10% left unchanged. The MLM loss is expressed in Equation 1, where x is an input token sequence drawn from domain distribution D, M is the set of randomly selected masked positions (|M|/|x| = 0.15), x_t is the original token at position t, x\M is the partially masked input, and theta represents the model parameters. The masked language modelling loss(L_MLM) is expressed mathematically in Equation (1) as

$$L_{MLM} = -E_{x\sim D} \sum_{t \in M} \log P\left(x_t \middle| x_{\setminus M}; \theta\right) \qquad (1)$$

The loss is minimised over 388,448 verses across 14 languages in the multilingual setting and 62,197 verses in the monolingual setting. Training used the AdamW optimiser [42] with a linear learning rate schedule peaking at 3e-5, 500 warmup steps, and weight decay 0.01. Effective batch size was 32 (16 per device with 2 gradient accumulation steps). Both models were trained for 30,000 steps with maximum sequence length of 128 tokens, using HuggingFace Transformers 5.0.0 with FP16 mixed precision on NVIDIA T4 and A100 GPUs via Google Colab. A custom checkpoint management callback using Python's shutil.rmtree permanently deleted all but the two most recent checkpoint directories after each evaluation step, preventing the accumulation of 211 GB of checkpoint files that were observed before this measure was implemented. All trained models were pushed to HuggingFace Hub immediately after training. Table 5 summarises the hyperparameter configuration.

Table 5. Hyperparameter configuration for continued pre-training

Parameter	**Value**
Training steps	30,000
Effective batch size	32 (16 per device, 2 gradient accumulation steps)
Learning rate	3e-5 (linear schedule, 500 warmup steps)
Weight decay	0.01 (AdamW optimiser [42])
MLM masking probability	15%
Masking strategy	80% [MASK], 10% random token, 10% unchanged

Parameter	**Value**
Max sequence length	128 tokens
Precision	FP16 mixed precision
Hardware	NVIDIA T4 and A100 GPUs (Google Colab)
Framework	HuggingFace Transformers 5.0.0
Checkpoints retained	2 (via PermanentDeleteCallback)

Figure 3 shows the full experimental pipeline across all five research objectives. Figures 4 and 5 show the WandB training dashboards and validation loss curves for both models. ReligionBERT's validation loss declined from 1.806 at step 500 to a best of 1.129 at step 28,500, a total reduction of 37.5%. MultiReligionBERT reached a best validation loss of 1.368 at step 29,500. The higher absolute loss for the multilingual model is expected given the greater linguistic diversity of its 12-language training data. Table 6 reports selected validation loss values.

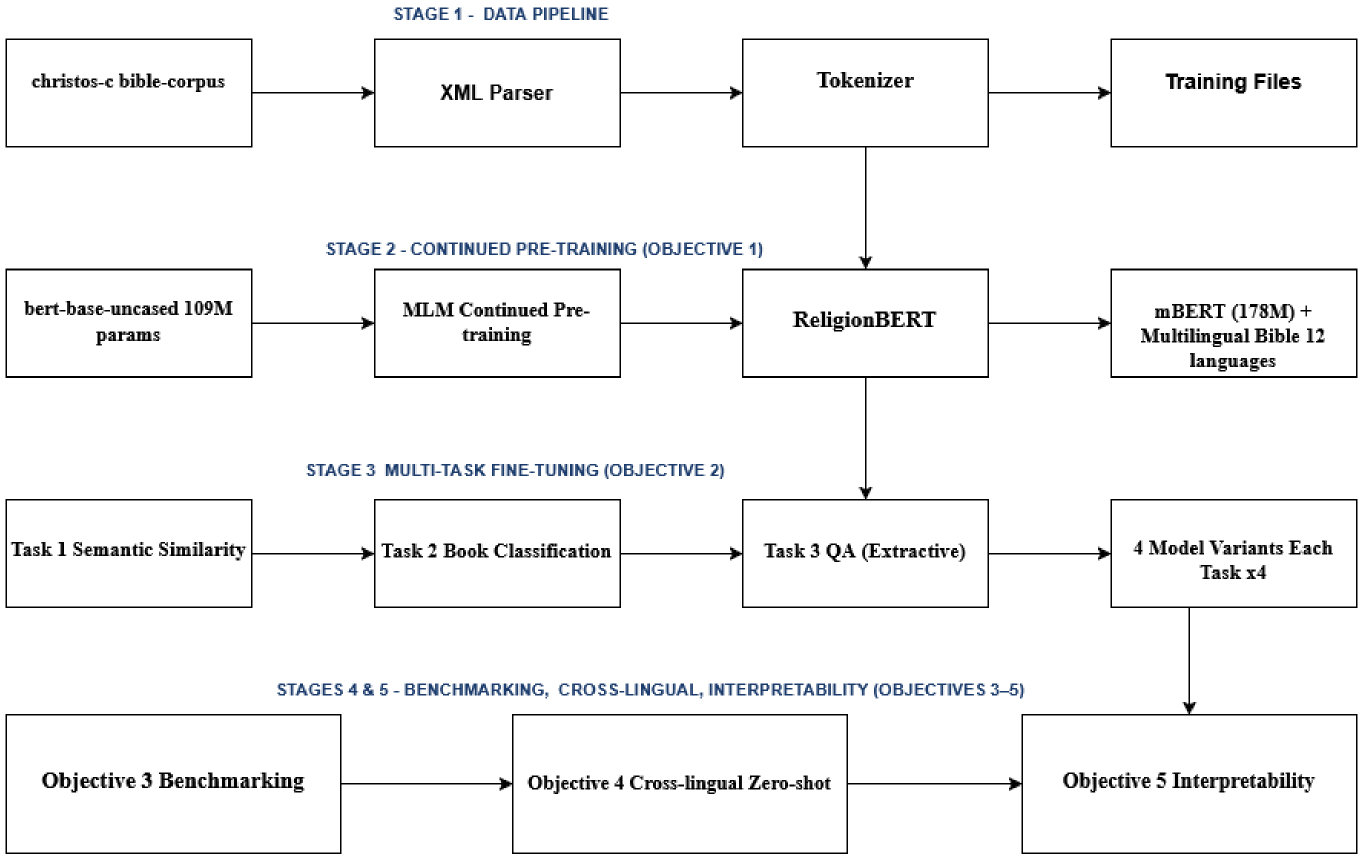

Figure 3. Full experimental pipeline from corpus collection through domain-adaptive pre-training, multi-task fine-tuning, benchmarking, cross-lingual evaluation, and interpretability analysis.

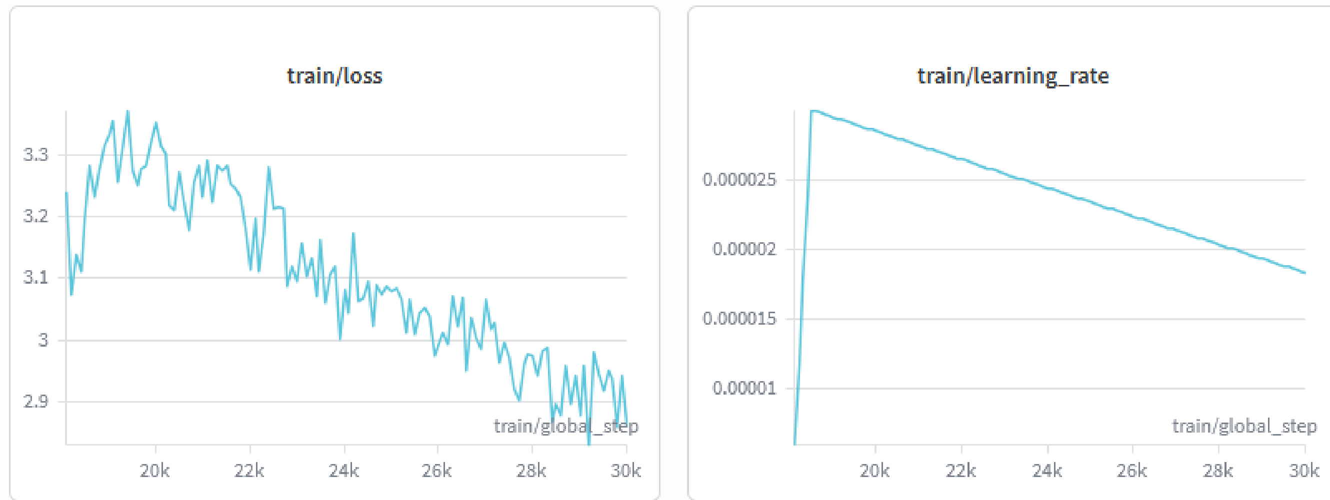

Fig. 4. WandB training dashboard showing ReligionBERT training loss and learning rate schedule across all 30,000 steps.

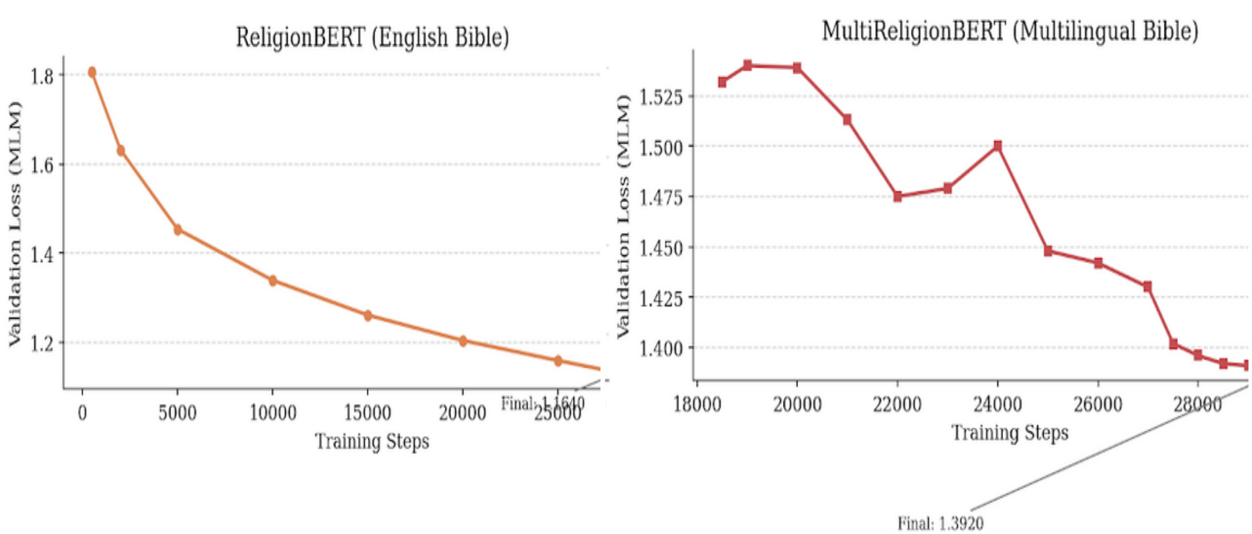

Fig. 5. Validation MLM loss curves. Left: ReligionBERT (final loss 1.164). Right: MultiReligionBERT (final loss 1.392).

Table 6. Selected validation loss values across training.

Model	Step 500	Step 10,000	Step 20,000	Best Loss	Best Step	Final
ReligionBERT	1.806	1.339	1.204	1.129	28,500	1.164
MultiReligionBERT	1.532*	n/a*	n/a*	1.368	29,500	1.392

* WandB logging for MultiReligionBERT began at step 18,500 following session interruptions.

3.6 Fine-Tuning Procedure

All four model variants were fine-tuned independently on each of the three tasks, producing twelve trained models in total. Figures 6 illustrate the two-stage training procedure.

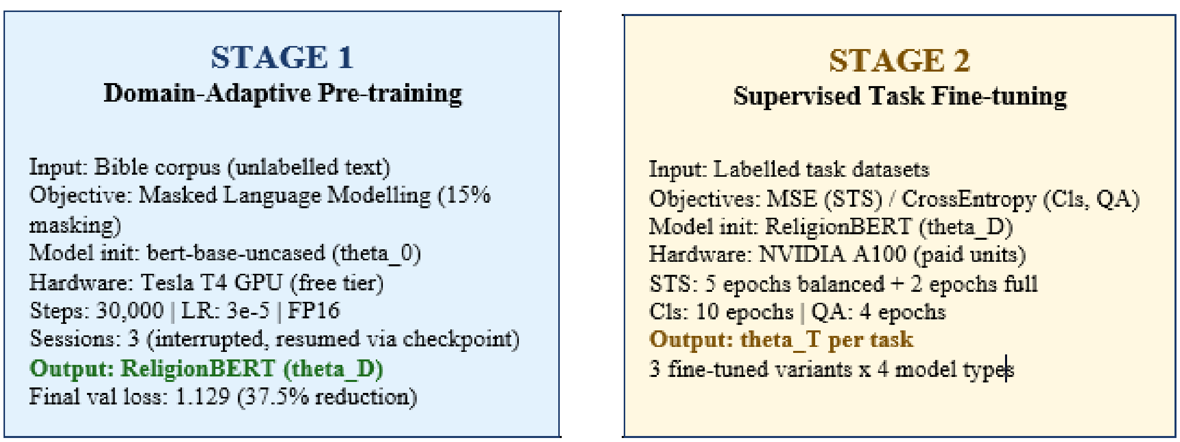

Fig. 6. Two-stage training procedure. Left: Stage 1 domain-adaptive pre-training. Right: Stage 2 supervised task fine-tuning task.

3.6.1 Semantic Similarity Fine-Tuning.

BertForSequenceClassification with num_labels=1 and mean squared error loss was used, producing a single continuous similarity score for each verse pair. A staged training strategy was applied: Stage 1 trained for 5 epochs on the balanced 6,392-pair subset at a learning rate of 2e-5 and batch size 32, providing an initial grounding in the full similarity range. Stage 2 continued training for 2 additional epochs on the full 21,994-pair dataset at a reduced learning rate of 1e-5, refining performance on the complete distribution. Evaluation used Pearson and Spearman correlation with gold similarity scores on the held-out test set.

3.6.2 Book Classification Fine-Tuning.

BertForSequenceClassification with num_labels=66 and cross-entropy loss was trained for 10 epochs at a learning rate of 2e-5 with batch size 32. Macro-averaged F1 was adopted as the primary evaluation metric to ensure that all 66 books are treated equally in the aggregate score, regardless of per-class sample count, which varies considerably across books.

3.5.3 Extractive QA Fine-Tuning.

BertForQuestionAnswering was used to predict start and end token positions, with input tokens tokenised at maximum length of 384 and a stride of 128 to handle verses and questions that together exceed the 512-token limit. Initial runs with 5 epochs at a learning rate of 2e-5 and batch size 16 produced severe overfitting after the first training epoch across all four model variants, attributable to the small training set of 959 examples. Hyperparameters were adjusted to 4 epochs, learning rate 1e-5, batch size 8, and warmup steps 50, which produced stable training curves without overfitting. Evaluation used standard SQuAD-style token-level F1 and Exact Match (EM) computed on the test set.

3.7.0 Evaluation of Model Performance.

Evaluation Metrics are critical for measuring and monitoring the performance of machine learning models, including the fine-tuned and adapted BERTs. The following metrics were adopted in the testing and evaluation of the models for verifying their performance after fine-tuning and baseline comparisons. The mathematical equations of the metrics are represented in Eq. (2–10) below.

Equation (2) is the expression for F1_macro that computes the unweighted mean F1 across all K classes, where K = 66, corresponding to the 66 books of the Bible. Macro F1 treats each class equally, regardless of its sample count and is preferred over weighted F1 for this task because the deliberate balanced sampling strategy makes equal treatment of classes scientifically appropriate.

EM is the Exact Match score, where $1[\cdot]$ is the indicator function that returns 1 when the normalised predicted answer string exactly matches the normalised gold answer string, and 0 otherwise, and N is the total number of evaluation examples as mathematically expressed in Equation (3). Both predicted and gold answers are normalised by lowercasing, removing punctuation, and stripping articles before comparison.

F1_token measures token-level overlap between the predicted and gold answer spans, where pred_tokens and gold_tokens are the sets of tokens in the predicted and gold answers, respectively, and the numerator counts tokens shared between both sets, depicted in Equation (5). Token F1 is more lenient than Exact Match and captures partial credit for answers that overlap with but do not exactly reproduce the gold answer string.

$$F1_k = \frac{2 \cdot Precision_k \cdot Recall_k}{Precision_k + Recall_k} \tag{2}$$

$$F1_{macro} = \frac{1}{K}\sum_{k=1}^{K} F1_k \tag{3}$$

$$EM = \frac{1}{N}\sum_{i=1}^{N}[normalize(pred_i) = normalize(gold_i)] \tag{4}$$

$$F1_{token} = \frac{2 \cdot |pred\_tokens \cap gold\_tokens|}{|pred\_tokens| + |gold\_tokens|} \tag{5}$$

where Precision_k = TP_k / (TP_k + FP_k) and Recall_k = TP_k / (TP_k + FN_k) for class k, and TP_k, FP_k, and FN_k are the true positives, false positives, and false negatives for class k, respectively, in Equation (2).

The following techniques were also utilized, either fully or in other related variants; they are three standard classification metrics widely adopted in natural language processing research.

Precision measures the proportion of correctly identified positive samples out of all samples predicted as positive. High precision indicates that the model produces few false positive predictions:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100 \tag{6}$$

where TP and FP represent true positive and false positive predictions respectively.

Accuracy represents the overall proportion of correctly classified samples across all classes:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100 \tag{7}$$

where TN represents true negative predictions. While accuracy provides a general indication of model performance, it can be misleading in class-imbalanced settings where a model may achieve high accuracy by predicting the dominant class.

F1-Score is the harmonic mean of precision and recall, providing a balanced measure that penalises models which sacrifice one for the other:

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \times 100 \tag{8}$$

Weighted F1-score is adopted in this study to account for class imbalance by weighting each class's F1 by its support in the test set, ensuring that performance on larger classes contributes proportionally to the overall score.

ρ in Equation (9) is the Spearman rank correlation, d_i is the rank difference for observation i, and n is the number of test pairs. Spearman measures monotonic rank agreement rather than linear agreement and is more robust to outliers and non-linear score distributions. Both metrics are reported together as they capture complementary aspects of model performance on the semantic similarity task.

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)} \qquad \textit{Eqn. (9)}$$

Perplexity was computed by applying 15% random masking and exponentiating the average per-masked-token cross-entropy loss, as expressed in Equation 2, where N is the number of evaluation sequences, M_i is the set of masked positions in sequence i, and x_t^(i) is the original token at position t.

$$PPL = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\frac{1}{|M_i|\sum_{t\in M_i}\log P\left(x_t^{(i)}\middle|x_{\setminus M_i}^{(i)};\theta\right)}\right) \tag{10}$$

4. Results and Discussion

4.1 Perplexity on Held-Out Religious Text

As a direct measure of domain alignment independent of any task-specific fine-tuning, perplexity was computed for generic BERT and ReligionBERT on a held-out set of 500 English Bible verses not included in pre-training.

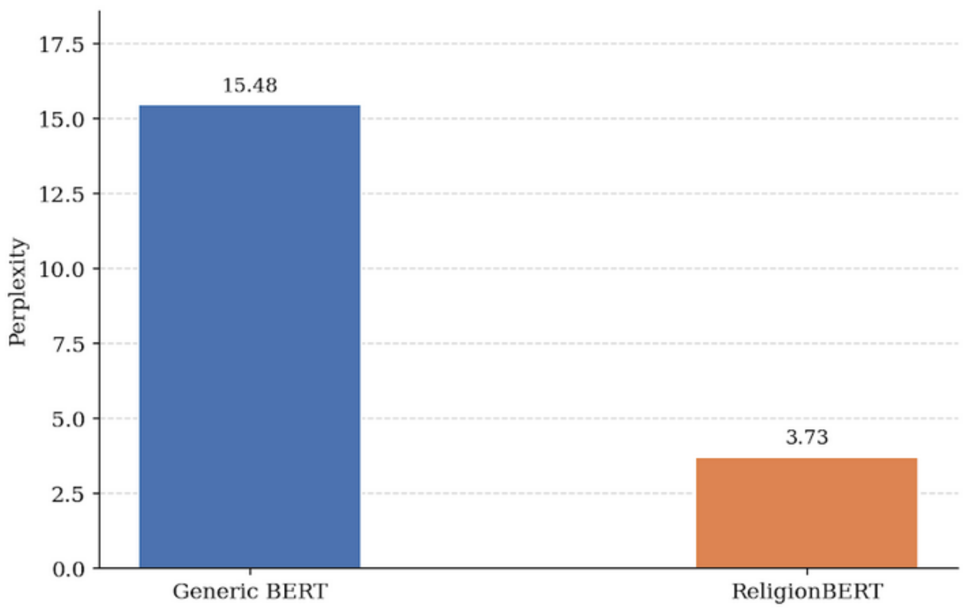

Fig. 7. Perplexity on 500 held-out English Bible verses (lower is better). ReligionBERT (3.73) vs. generic BERT (15.48)

ReligionBERT achieves a perplexity of 3.73 compared to 15.48 for generic BERT, a reduction of 75.9%. This result confirms that continued pre-training on the Bible corpus substantially improves the model's ability to predict masked tokens in religious text, demonstrating genuine domain alignment independent of any downstream task performance. A perplexity reduction of this magnitude is consistent with the gains reported by BioBERT and SciBERT on their respective target domains and provides strong a priori evidence that the domain adaptation is effective before any fine-tuning is applied.

4.2 Semantic Similarity

Table 7 and Figure 8 report test set results on the semantic similarity task. ReligionBERT outperforms generic BERT on both Pearson (+0.0054) and Spearman (+0.0035) correlation. MultiReligionBERT outperforms mBERT on Pearson by 0.0011 but shows a slight Spearman decrease of 0.0103. The high base performance above 0.95 Pearson correlation across all models reflects a ceiling effect inherent in the dataset design: the dominant class consists of cross-translation pairs of identical verses, and any BERT variant fine-tuned on a sufficient sample of such pairs converges to very high Pearson correlation. The Spearman metric is more sensitive to performance differences because it captures rank-order agreement and is more responsive to errors in the middle and low similarity ranges.

Table 7. Semantic similarity test set results.

Model	Pearson	Spearman	vs. Baseline (Pearson)	vs. Baseline (Spearman)
Generic BERT	0.9569	0.6556	Baseline	baseline
ReligionBERT	0.9623	0.6591	+0.0054	+0.0035
mBERT	0.9624	0.6772	Baseline	Baseline
MultiReligionBERT	0.9635	0.6669	+0.0011	-0.0103

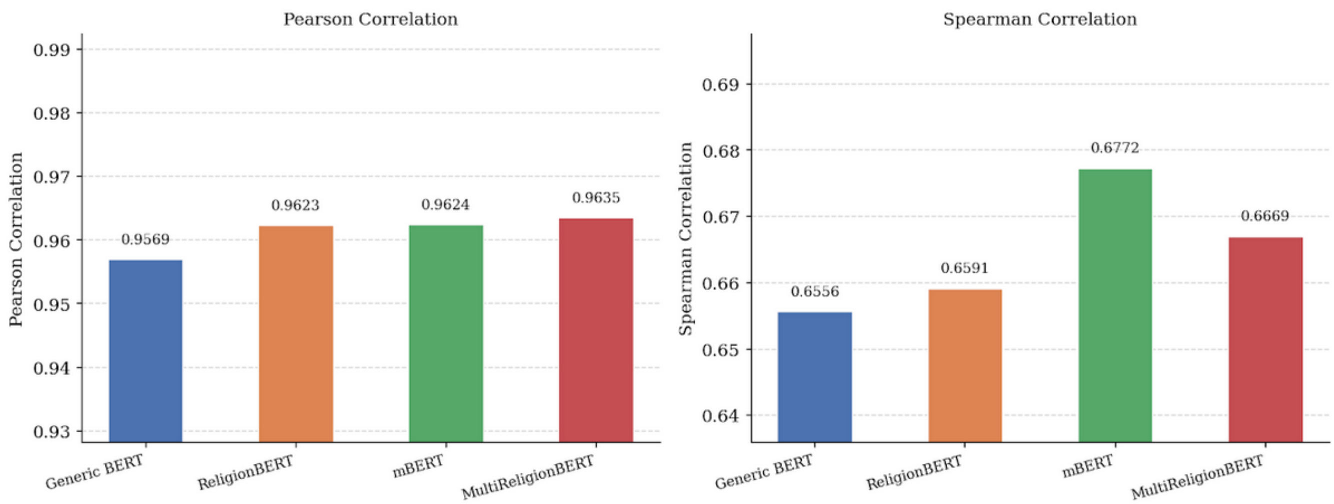

Fig. 8. Pearson (left) and Spearman (right) correlation for all four models on the semantic similarity test set

4.3 Book Classification

Table 8 and Figure 9 report results on the 66-class book classification task. Random chance yields 1.5% accuracy across 66 equally probable classes. All models substantially exceed this baseline. ReligionBERT outperforms generic BERT by 2.72 accuracy points and 2.37 macro F1 points. MultiReligionBERT outperforms mBERT by 3.88 accuracy points and 4.21 macro F1 points. The larger gain for the multilingual pair likely reflects the stronger domain alignment signal available from 12 languages of biblical text simultaneously, which provides more varied contextual representations of the same theological concepts across different linguistic encodings. The book classification task is particularly demanding because it requires the model to distinguish between 66 stylistically and thematically distinct classes based on a single verse, without any cross-verse context.

Table 8. Book classification test set results.

Model	Accuracy	Macro F1	Acc vs. Baseline	F1 vs. Baseline
Generic BERT	0.4075	0.3144	baseline	Baseline
ReligionBERT	0.4347	0.3381	+0.0272	+0.0237
mBERT	0.3972	0.2948	baseline	Baseline
MultiReligionBERT	0.4360	0.3369	+0.0388	+0.0421

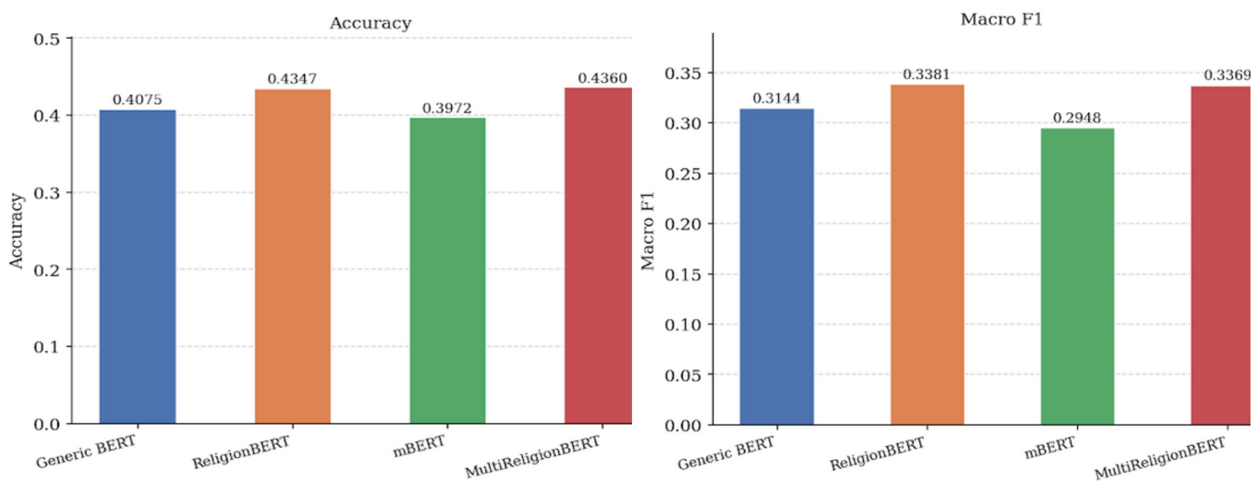

Fig. 9. Accuracy (left) and Macro F1 (right) for all four models on the 66-class book classification test set.

4.4 Extractive Question Answering

Table 9 and Figure 10 report results on the extractive QA test set. ReligionBERT demonstrates the largest single gain from domain adaptation across all three tasks, improving over generic BERT by 7.50 Exact Match points and 1.57 Token F1 points. This result is consistent with the nature of extractive QA: precise span selection is highly sensitive to token-level contextual alignment, and domain-specific pre-training provides exactly that grounding for religious passages. The model that has learned to assign contextually appropriate representations to theologically significant tokens is better positioned to identify the precise character offset of an answer span within a verse. MultiReligionBERT shows an unexpected reversal, scoring 7.50 EM points below mBERT.

This counterintuitive result is best understood as a manifestation of the known risk of catastrophic forgetting during continued pre-training: by adapting strongly to the multilingual religious domain across 12 languages, MultiReligionBERT shifts its English-language representations in ways that impair English-specific span extraction on a small, English-only QA dataset. The effect is more pronounced for MultiReligionBERT than for ReligionBERT because the multilingual setting involves a more aggressive redistribution of representational capacity across languages, leaving less dedicated capacity for English fine-grained span matching.

Table 9. Extractive QA test set results (Exact Match % and Token F1 %).

Model	Exact Match (%)	Token F1 (%)	EM vs. Baseline	F1 vs. Baseline
Generic BERT	32.50	58.91	baseline	Baseline
ReligionBERT	40.00	60.48	+7.50	+1.57
mBERT	48.33	70.54	baseline	Baseline
MultiReligionBERT	40.83	65.39	-7.50	-5.15

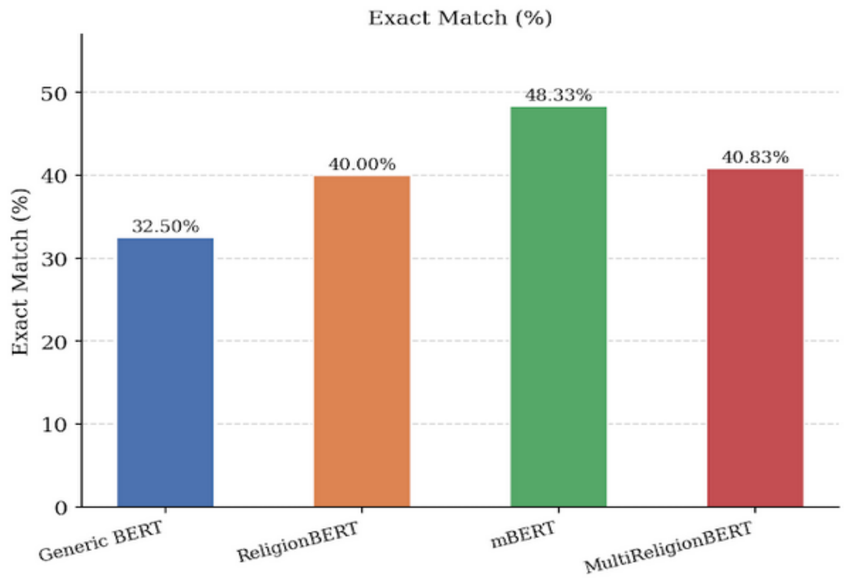

Fig. 10a

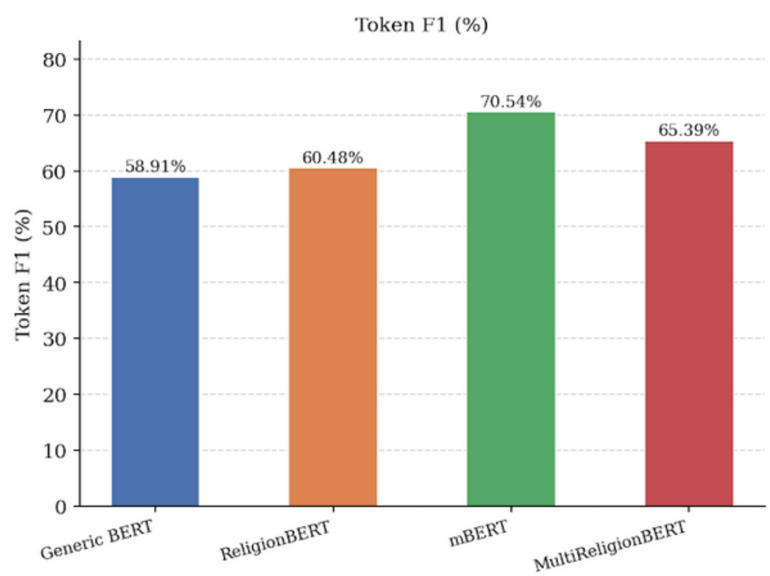

Fig. 10b

Fig. 10. Exact Match (**Fig. 10a**) and Token F1 (**Fig. 10b**) for all four models on the extractive QA test set.

4.5 Benchmarking Synthesis

Table 10 consolidates all results across the three tasks and four models. ReligionBERT outperforms generic BERT on five of six evaluation metrics. MultiReligionBERT outperforms mBERT on four of six metrics. The domain adaptation advantage is strongest on extractive QA for the English model (+7.50 EM) and on book classification for the multilingual model (+3.88 accuracy points). The weakest gain is on semantic similarity Pearson correlation, where the ceiling effect limits observable differences. Figures 11, 12, and 13 provide visual syntheses of the results across all tasks simultaneously.

Table 10. Complete benchmarking results across all tasks and models.

Model	Sim Pearson	Sim Spearman	Cls Acc	Cls F1	QA EM%	QA F1%
Generic BERT	0.9569	0.6556	0.4075	0.3144	32.50	58.91
ReligionBERT	0.9623	0.6591	0.4347	0.3381	40.00	60.48
mBERT	0.9624	0.6772	0.3972	0.2948	48.33	70.54
MultiReligionBERT	0.9635	0.6669	0.4360	0.3369	40.83	65.39

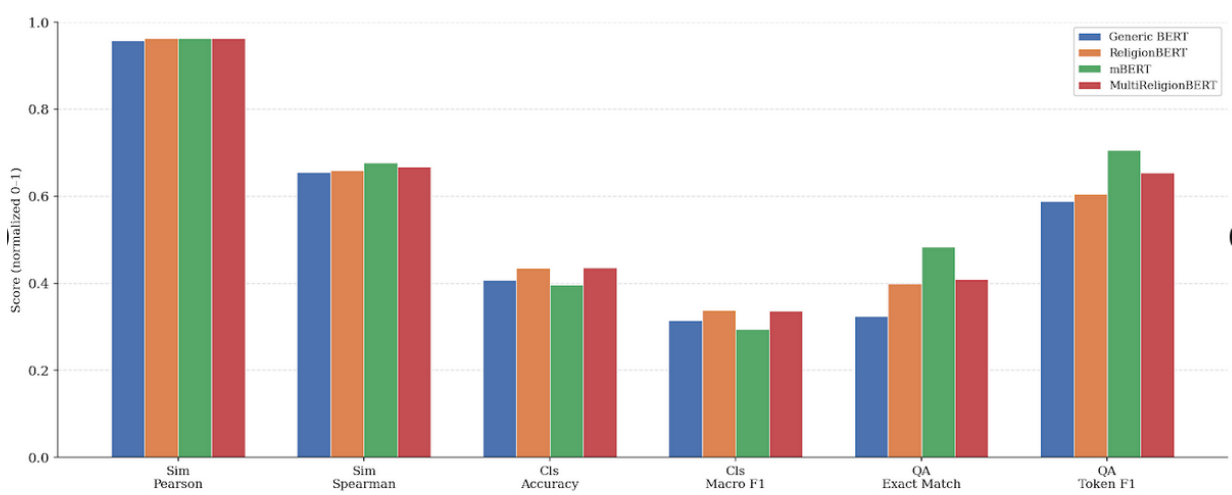

Fig. 11. All four models across all six evaluation metrics on a normalised 0-1 scale.

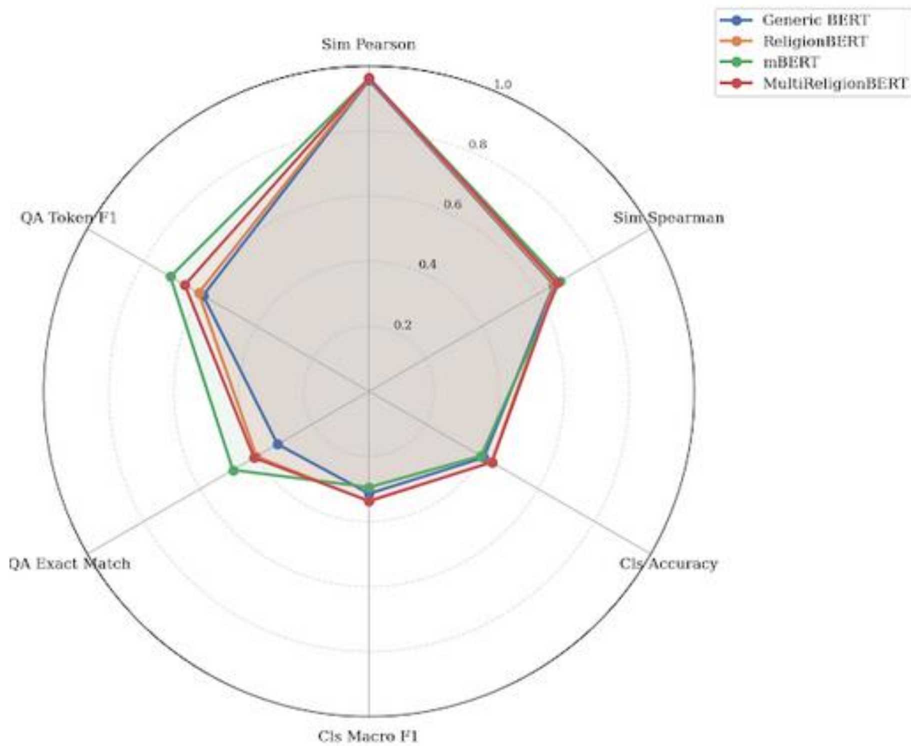

Fig. 12 Radar chart across all six metrics.

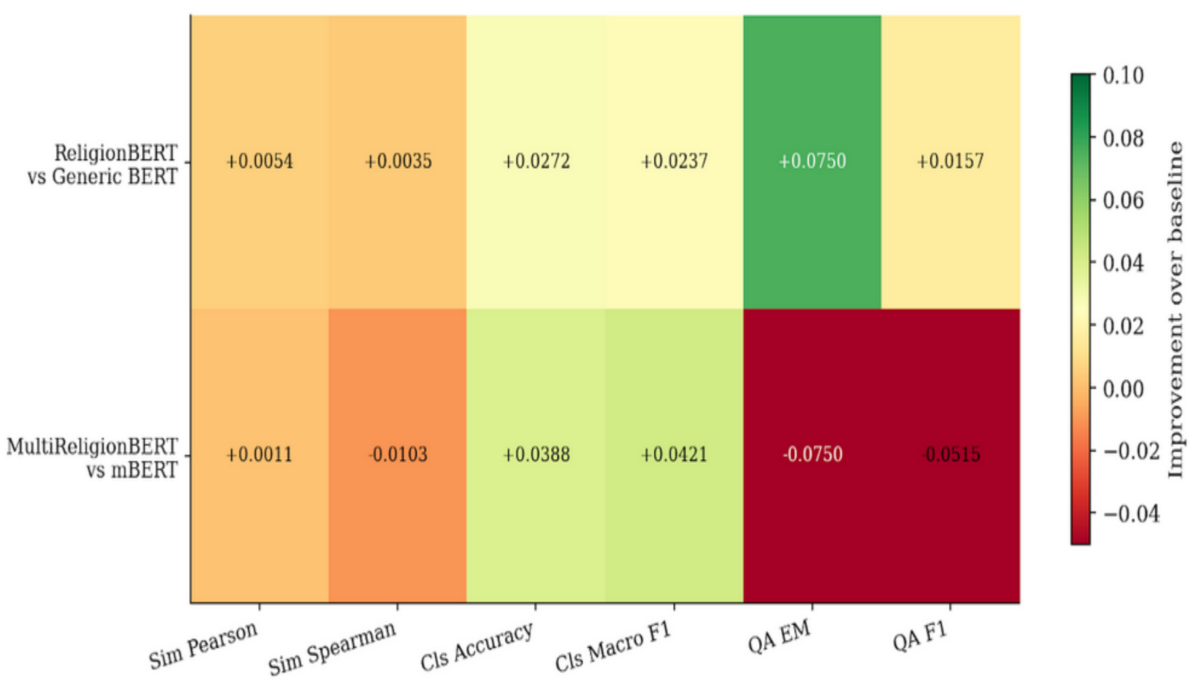

Fig. 13 Improvement heatmap of domain-adapted vs. generic models. Green = gain; red = loss

4.6 Cross-Lingual Zero-Shot Transfer

Three multilingual models, mBERT, MultiReligionBERT, and XLM-RoBERTa [18], were evaluated on zero-shot book classification applied to five African language Bible corpora: Amharic (Semitic family), Shona, Xhosa, and Swahili (Bantu family), and Ewe (Gbe family). All models were fine-tuned exclusively on English classification data; no target-language fine-tuning was performed. XLM-RoBERTa was fine-tuned on the English training set first, reaching 35.83% test accuracy and 0.2617 macro F1. Evaluation sets were constructed by sampling approximately four verses per available book in each target language. Table 11 reports full results.

Table 11. Zero-shot cross-lingual book classification results on five African languages.

Language	Family	Samples	mBERT Acc	MultiRel Acc	XLM-R Acc	mBERT F1	MultiRel F1	XLM-R F1
Amharic	Semitic	264	0.0152	0.0152	0.1326	0.0005	0.0005	0.0886
Shona	Bantu	252	0.0556	0.0714	0.0556	0.0267	0.0402	0.0381
Xhosa	Bantu	264	0.0189	0.0227	0.0530	0.0040	0.0072	0.0396
Ewe	Gbe	297	0.0000	0.0337	0.0034	0.0000	0.0187	0.0011
Swahili	Bantu	286	0.0280	0.0559	0.0874	0.0098	0.0262	0.0364

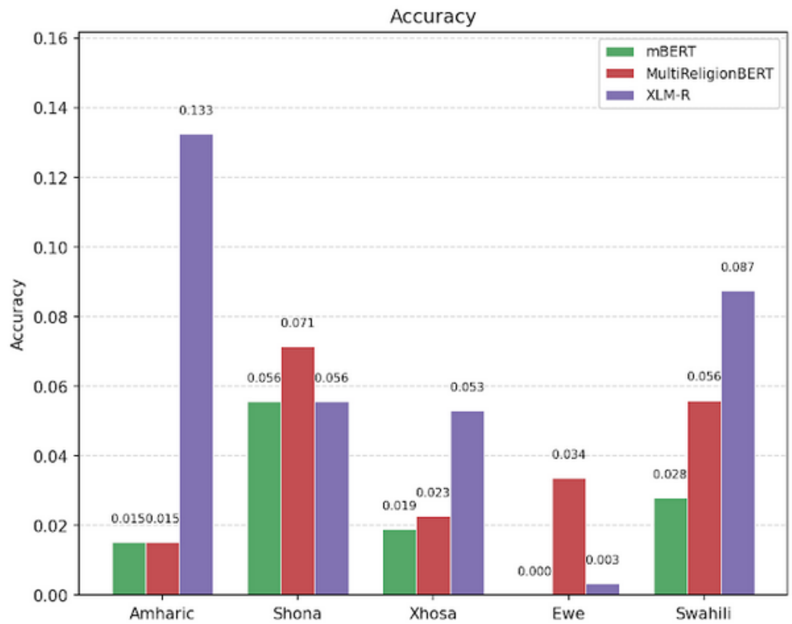

Fig. 14. Zero-shot accuracy for mBERT (green), MultiReligionBERT (red), and XLM-RoBERTa (blue) across five African languages.

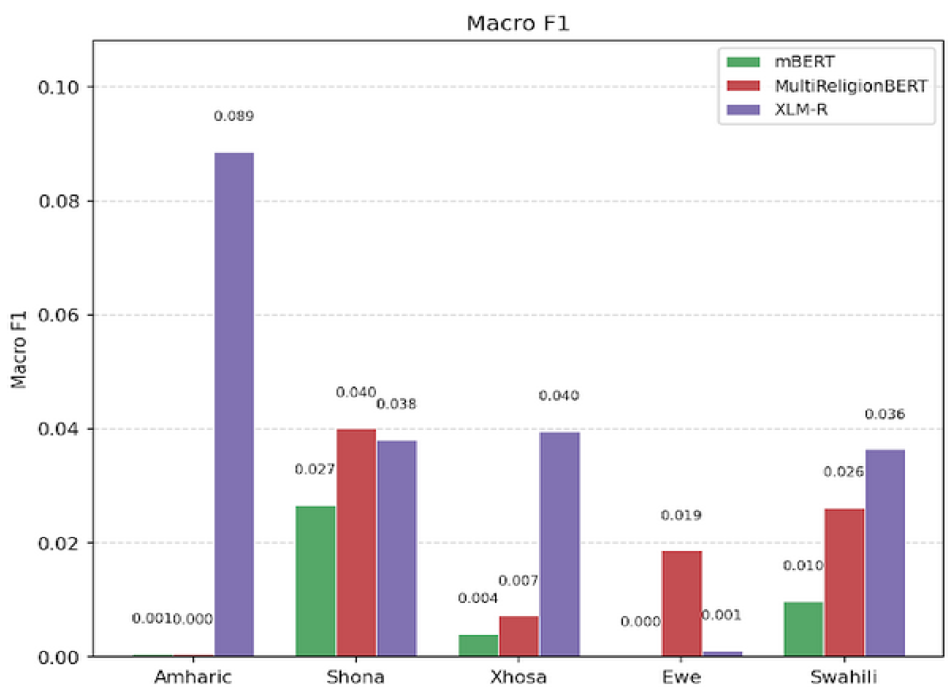

Fig. 15 Macro F1 across five African languages

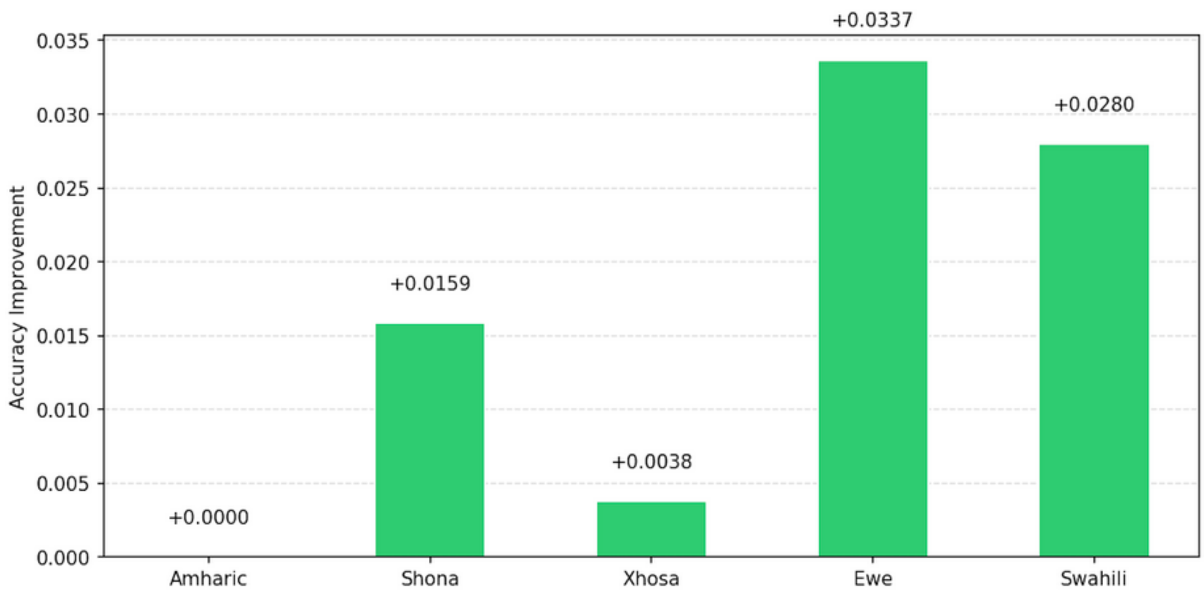

Fig. 16 MultiReligionBERT accuracy gain over mBERT per language.

MultiReligionBERT outperforms mBERT on four of five languages. The most striking result is Ewe: mBERT predicts zero correct book labels across all 297 test samples while MultiReligionBERT achieves 3.37% accuracy. Ewe is a severely low-resource language with minimal representation in Wikipedia and CommonCrawl, the corpora underlying most multilingual models. Its inclusion in the Bible pre-training data, through a New Testament translation, provides MultiReligionBERT with transfer signal that mBERT's general multilingual pre-training cannot supply. XLM-RoBERTa leads on Amharic, Xhosa, and Swahili, consistent with its larger general pre-training corpus, but trails MultiReligionBERT on Ewe, where domain-specific religious corpus coverage provides an advantage that general-purpose scale alone does not match.

The cross-lingual results have implications that extend beyond this study. Bible translations are available in more than 2,000 languages, including a substantial proportion of sub-Saharan African languages for which no other consistent digital text resource exists. The demonstrated advantage of MultiReligionBERT over mBERT on Ewe suggests that leveraging Bible translations as pre-training data for low-resource African language NLP could provide meaningful benefits for a wide range of languages currently underserved by standard multilingual models. Future work should investigate whether this advantage holds for tasks beyond book classification, and whether increasing the number of African language Bible translations in the pre-training corpus improves cross-lingual transfer further.

4.7 Evidence of Genuine Semantic Adaptation

To assess whether the observed performance gains reflect genuine shifts in internal representations rather than surface-level vocabulary matching, linear probing classifiers were trained on frozen CLS token representations extracted from all 13 representational depths (embedding layer plus 12 transformer layers) of generic BERT and ReligionBERT. The probing classifiers were logistic regression models trained on three tasks: book classification (66 classes), testament classification (binary: Old Testament versus New Testament), and theological section classification (7 categories: Law, History, Poetry, Prophecy, Gospels, Epistles, and Apocalyptic).

Table 12 reports final-layer probing accuracy. ReligionBERT outperforms generic BERT at the final transformer layer on all three tasks, with the largest gain of 7.12 percentage points on theological section classification. This result indicates that domain adaptation meaningfully shifts the model's representational geometry toward religious categorical distinctions, providing mechanistic support for the claim that the downstream performance gains are not merely artefacts of vocabulary shift but reflect deeper semantic adaptation.

Table 12. Probing classifier accuracy at the final transformer layer (layer 12) for generic BERT and ReligionBERT.

Probing Task	Classes	Generic BERT	ReligionBERT	Absolute Gain
Book Classification	66	0.3493	0.4023	+0.0530
Testament (OT vs. NT)	2	0.8655	0.9017	+0.0362
Theological Section	7	0.6611	0.7322	+0.0712

4.8 The LLM-Assisted Dataset Pipeline as a Methodological Contribution

The extractive QA dataset construction pipeline represents a methodological contribution that is transferable beyond this study. For specialised domains in which no annotated benchmark exists, the combination of LLM-assisted annotation with structured prompting, automated substring validation, and human spot-checking provides a practical and cost-effective path to task-specific datasets. The 1,199-example dataset was generated at a total API cost under USD 5.00 and achieved 100% verified quality on a 200-sample human review.

The structured prompt format, validation logic, and dataset schema are fully documented and directly applicable to other sacred text traditions, including the Quran, the Bhagavad Gita, the Buddhist Pali Canon, and the Talmud, for none of which does a standardised extractive QA benchmark currently exist. This opens a systematic path toward multilingual, multi-tradition religious NLP benchmarking that the community has lacked.

5. Conclusion

This paper has presented ReligionBERT and MultiReligionBERT, two domain-adapted BERT models produced by continued masked language modelling pre-training on Biblical corpora in monolingual and multilingual settings respectively. The study provides the first systematic multi-task, multi-model benchmarking of Bible corpus domain adaptation in the NLP literature, covering three task types (semantic similarity, book classification, and extractive question answering), four model variants (generic BERT, ReligionBERT, mBERT, MultiReligionBERT), and five African languages for cross-lingual transfer evaluation.

The principal empirical findings are as follows. ReligionBERT outperforms generic BERT on five of six evaluation metrics, with the peak gain of +7.50 Exact Match points on extractive QA, demonstrating that domain-specific pre-training most benefits tasks that demand fine-grained token-level span matching in the target domain. Perplexity on held-out Bible text falls by 75.9% from generic BERT to ReligionBERT, providing a direct and task-independent confirmation of domain alignment.

MultiReligionBERT outperforms mBERT on four of six metrics and on four of five African languages in zero-shot cross-lingual transfer, with the Ewe result being the most striking: mBERT achieves 0.00% accuracy on all 297 Ewe test samples while MultiReligionBERT achieves 3.37%, demonstrating that inclusion of a language in the religious pre-training corpus provides a transfer signal that general multilingual pre-training cannot supply. Probing classifier experiments at all 13 representational depths confirm that the performance gains reflect genuine shifts in how the models internally encode religious categorical distinctions, with the largest probing gain of 7.12 percentage points on theological section classification providing mechanistic evidence of semantic adaptation.

Three methodological contributions accompany the empirical findings. The three automatically constructed datasets, totalling 21,994 semantic similarity pairs, 7,726 book classification samples, and 1,199 extractive QA examples, represent the first structured, openly available benchmarks for religious NLP of this kind. The LLM-assisted QA construction pipeline, achieving 100% verified quality at under USD 5.00 in API costs, is directly transferable to other sacred text traditions and opens a systematic path toward multi-tradition religious NLP benchmarking. The correction of eleven non-standard XML book code mappings in the christos-c/bible-corpus repository constitutes a practical data quality contribution that benefits any researcher using this widely accessed resource.

Several limitations of the present study define clear directions for future work. The pre-training corpus is restricted to the Bible, and religious NLP encompasses a much broader set of traditions, including the Quran, the Vedas, the Buddhist Pali Canon, and Talmudic literature, each with distinct linguistic properties, script systems, and intertextual structures. Extending the domain adaptation paradigm to these traditions would require separate corpus construction efforts but could leverage the LLM-assisted dataset construction pipeline developed here.

The extractive QA dataset, while verified at high quality, is modest in scale at 959 training examples, and the overfitting observed during initial training indicates that performance on this task is likely bottlenecked by dataset size rather than model capacity. Scaling the dataset using the documented pipeline to several thousand examples would likely produce further gains. The cross-lingual evaluation is limited to book classification; extending it to cross-lingual extractive QA or semantic similarity using multilingual verse pair datasets would provide a more complete picture of MultiReligionBERT's cross-lingual capabilities.

Both models, ReligionBERT and MultiReligionBERT, are publicly available on HuggingFace at LucasLicht/religion-bert and LucasLicht/multi-religion-bert respectively. All three fine-tuning datasets and the full QA generation pipeline, including the structured prompts, validation code, and dataset schema, are released alongside the models. We hope that these resources lower the barrier to entry for NLP research on religious and sacred texts, support computational scholarship across faith traditions, and contribute to the development of NLP tools for the many low-resource African and minority languages for which Bible translations represent the most consistent and systematically structured digital text resource available.

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The Groq API was used for LLM-assisted QA dataset generation. Computation was performed on NVIDIA T4 and A100 GPUs via Google Colab. Training metrics were tracked using Weights and Biases. Model weights are hosted on HuggingFace Hub.

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